WORKED SOLUTION

STEP III 2015, Question 5

Pure • Irrationality by Descent

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q5

This question has a full guided treatment in *Beyond STEP*, Section 1.3, *The Smallest Counterexample*: https://beyondstep.keithmcnulty.org/ch01/well-ordering.html

The key idea

This is proof by *infinite descent*: if $\sqrt{2}$ were rational there would be a smallest positive integer that turns $\sqrt{2}$ into an integer when multiplied by it, and we then build a smaller one. Multiplying by $\sqrt{2}-1$ does the job, because it is a number between 0 and 1 that keeps us inside the set. For the cube root of 2, the same trick needs a set that also keeps track of $2^{2/3}$, which is why the question first asks you to tie the two numbers together.

Solution

Part (i)

Step 3. By the assumption in step 1, $\sqrt{2} = \frac{p}{q}$ for some integers p and q with $q \neq 0$. Since $\sqrt{2} > 0$ we may take p and q both positive. Then

$$q\sqrt{2} = p$$

is an integer, and q is a positive integer, so $q \in S$. Hence S is not empty.

Step 5. Since $k \in S$, the number $m = k\sqrt{2}$ is an integer. We need to check two things about $n = (\sqrt{2}-1)k$.

- n is a positive integer: $n = k\sqrt{2} - k = m - k$ is a difference of integers, and $n > 0$ because $k > 0$ and $\sqrt{2} > 1$.
- $n\sqrt{2}$ is an integer:

  $$n\sqrt{2} = (\sqrt{2}-1)k\sqrt{2} = 2k - k\sqrt{2} = 2k - m,$$

  again a difference of integers.

So $(\sqrt{2}-1)k \in S$.

Step 6. Since $1 < \sqrt{2} < 2$ (as $1 < 2 < 4$), we have $0 < \sqrt{2}-1 < 1$, and so, as $k > 0$,

$$(\sqrt{2}-1)k < k.$$

Step 5 therefore gives a positive integer in S that is smaller than k, contradicting the choice of k in step 4 as the smallest element of S. (Step 4 is legitimate, since every non-empty set of positive integers has a smallest element, and step 3 showed S is non-empty.) The only assumption made was step 1, so it must be false: $\sqrt{2}$ is irrational.

Part (ii)

Write $a = 2^{1/3}$, so $a^2 = 2^{2/3}$ and $a^3 = 2$.

The equivalence. If a is rational then so is a^2, a product of rationals. Conversely, if a^2 is rational, then

$$a = \frac{a^3}{a^2} = \frac{2}{a^2}$$

is rational, since $a^2 \neq 0$. So $2^{1/3}$ is rational if and only if $2^{2/3}$ is.

Irrationality. Suppose, for contradiction, that a is rational. By the equivalence, a^2 is rational as well. Define T to be the set of positive integers n such that *both* na and na^2 are integers.

> **?** **Ask yourself** Why not just copy part (i) with $S = \{n : na \text{ is an integer}\}$?

In part (i) the check that $(\sqrt{2}-1)k \in S$ needed $k(\sqrt{2})^2 = 2k$ to be an integer, which is automatic. Here, multiplying $(a-1)k$ by a gives $ka^2 - ka$, and nothing in the condition "ka is an integer" says that ka^2 is an integer. So we build that into the set, and this is where we need the first half of the part.

T is non-empty. Write $a = \frac{p}{q}$ with p, q positive integers. Then $q^2 a = pq$ and $q^2 a^2 = p^2$ are integers, so $q^2 \in T$.

Descent. Let k be the smallest element of T, so $u = ka$ and $v = ka^2$ are integers. Consider $n = (a-1)k$.

- $n = u - k$ is an integer, and $n > 0$ since $a > 1$ (as $1^3 < 2$).
- $na = ka^2 - ka = v - u$ is an integer.
- $na^2 = ka^3 - ka^2 = 2k - v$ is an integer, using $a^3 = 2$.

So $n \in T$. But $a < 2$ (as $2^3 = 8 > 2$), so $0 < a - 1 < 1$ and $n = (a-1)k < k$, contradicting the minimality of k. Hence $2^{1/3}$ is irrational, and by the equivalence $2^{2/3}$ is irrational too.

Checks

- Try the descent on a near miss. The fraction $\frac{7}{5}$ is close to $\sqrt{2}$; the map $(k, m) \mapsto (m-k, 2k-m)$ from step 5 sends $(5,7)$ to $(2,3)$, and $3^2 - 2\cdot 2^2 = 1 = -(7^2 - 2\cdot 5^2)$. The quantity $m^2 - 2k^2$ only changes sign, so if it were 0 (that is, $m = k\sqrt{2}$) it would stay 0 while k kept shrinking, which is impossible.
- Numerically, $\sqrt{2} - 1 \approx 0.414$ and $2^{1/3} - 1 \approx 0.260$, both strictly between 0 and 1, as the descent needs.
- A computer search over all $n \leq 2 \times 10^5$ finds no n with $2n^2$ a perfect square, or with $2n^3$ or $4n^3$ a perfect cube.

Common slips

- Forgetting to show that $(\sqrt{2}-1)k$ is *positive*: S contains only positive integers, so this is needed.
- In part (ii), using a set defined by na alone and then asserting without reason that ka^2 is an integer.
- Not saying why the smallest element k exists: it needs S (or T) to be non-empty, which is the point of step 3.

Beyond STEP

The argument is really about the well-ordering of the positive integers, the principle behind induction, and it generalises: if $\sqrt[r]{N}$ is not an integer then it is irrational, as one sees by taking the set of n for which $n\alpha, n\alpha^2, \ldots, n\alpha^{r-1}$ are all integers and multiplying its least element by $\alpha - \lfloor \alpha \rfloor$, which lies strictly between 0 and 1. In a first course in algebra this becomes the statement that the integers form an *integrally closed* ring: a rational root of a monic polynomial with integer coefficients must be an integer. Davenport's *The Higher Arithmetic* is a lovely place to read more about the integers and their properties.

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